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Phase-Inversion Polymer Composite Separators Based on Hexagonal Boron Nitride Nanosheets for High-Temperature Lithium-Ion Batteries

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Abstract

By preventing electrical contact between anode and cathode electrodes while promoting ionic transport, separators are critical components in the safe operation of rechargeable battery technologies. However, traditional polymer-based separators have limited thermal stability, which has contributed to catastrophic thermal runaway failure modes that have conspicuously plagued lithium-ion batteries. Here, we describe the development of phase-inversion composite separators based on carbon-coated hexagonal boron nitride (hBN) nanosheets and polyvinylidene fluoride (PVDF) polymers that possess high porosity, electrolyte wettability, and thermal stability. The carbon-coated hBN nanosheets are obtained through a scalable liquid-phase shear exfoliation method using ethyl cellulose as a polymer stabilizer and source of the carbon coating following thermal pyrolysis. When incorporated within the PVDF matrix, the carbon-

coated hBN nanosheets promote favorable interfacial interactions during the phase-inversion process, resulting in porous, flexible, free-standing composite separators. The unique chemical composition of these carbon-coated hBN separators implies high wettability for a wide range of liquid electrolytes. This combination of high porosity and electrolyte wettability enables enhanced ionic conductivity and lithium-ion battery electrochemical performance that exceeds incumbent polyolefin separators over a wide range of operating conditions. The hBN nanosheets also impart high thermal stability, providing safe lithium-ion battery operation up to 120 °C.

Introduction

The high energy density and lightweight design of rechargeable lithium-ion batteries (LIBs) have enabled the rapid ascent of portable electronics, electric vehicles, wireless power tools, and large-scale energy storage systems.¹⁻³ LIBs are sophisticated electrochemical systems that rely on transport of lithium ions between the anode and cathode through an ionically conductive electrolyte. A key component of a LIB is the electrically insulating separator, which acts as a physical barrier between the two electrodes by preventing electrical contact while also promoting ion transport during charge and discharge via pathways for ionic conduction throughout an interconnected porous structure.^{4,5}

Since the separator has a crucial role in ion transport, it influences LIB performance, cycle life, and safety.⁶ For optimal LIB operation, the separator needs to meet several requirements including: (1) high chemical and electrochemical stability to resist degradation due to reactivity with the electrolyte and electrodes under strong oxidizing and reducing conditions; (2) high porosity to provide interconnected pathways for efficient ion transport; (3) high wettability to absorb and hold a large amount of liquid electrolyte

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3 during cell operation; (4) thermal resistance to ensure safe operation at increased
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7 temperatures and during local temperature changes.^{7,8}
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10 Commercial manufacturing of LIBs utilizes polyolefin microporous separators based on
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13 polypropylene (PP) and polyethylene (PE).⁸ Despite their high chemical and mechanical
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17 stability and relatively low cost, these conventional polymer membranes have inherent
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21 drawbacks including low wettability for liquid electrolytes, which imposes limits on the
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25 performance of LIB cells.⁹ In addition, polyolefin separators possess relatively low glass
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28 transition temperatures that result in shrinkage and compromised mechanical integrity,
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31 which ultimately lead to internal electrical shorting and catastrophic LIB failure modes at
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35 elevated temperatures.¹⁰⁻¹²
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38 Considerable research has been devoted to overcoming these challenges with the aim
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41 of improving the performance and safety of separators.^{13,14} For instance, the intrinsic
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45 hydrophobicity of polyolefins has been modified by grafting hydrophilic monomers or
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49 functional groups on the membrane surfaces, achieving improved hydrophilicity for
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53 efficient electrolyte wetting.¹⁵⁻¹⁷ Alternatively, ceramic particle coatings (e.g., SiO₂ or
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56 Al₂O₃) on the surface of polyolefin membranes lead to improved electrolyte wettability and
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3 thermal stability.¹⁸⁻²⁰ However, these ceramic particle coatings require the use of organic
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7 binders that still limit the maximum safe operating temperature while also lowering
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10 membrane porosity. Moreover, due to relatively weak interactions with the polymer host
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13 and membrane swelling upon exposure to the electrolyte, coating delamination often
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16 occurs during electrochemical cycling, which limits battery cycle life.²¹ Non-woven
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19 separators based on natural and synthetic fibers have also been reported including the
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22 use of electrospun thermosetting polymers that provide improved thermal stability,²² but
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25 these structures lack uniform pore size distribution due to macroscale porosity.^{10,12,13}
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31 In addition to ceramic particle coatings on conventional polymer separators, composite
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34 membranes have been produced by direct incorporation of inorganic particles such as
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37 SiO₂, Al₂O₃, NiO, MgO, and TiO₂ into a polymer matrix to promote enhanced mechanical
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40 strength, wettability, ionic conductivity, and thermal resistance.^{5,13,23,24} More recently,
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43 hexagonal boron nitride (hBN), which is an electrically insulating isomorph of graphene,
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46 has drawn attention as a ceramic filler because of its high thermal conductivity, chemical
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49 inertness, and mechanical robustness.²⁵ Moreover, hBN is free of dangling bonds or
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52 surface charge traps.^{26,27} In the context of battery separators, hBN has been incorporated
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3 into polymer matrices such as polyethylene and polyvinylidene fluoride-
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7 hexafluoropropylene to create composite separators with improved thermal homogeneity
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10 that promotes uniform nucleation and growth of lithium metal while enhancing
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13 electrochemical performance.^{28,29} hBN has also been coated on the surface of
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17 commercial polyolefin separators to increase the stability of lithium metal anodes.³⁰
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21 Similar to other 2D materials, large quantities of hBN nanosheets can be produced via
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24 liquid-phase exfoliation of bulk crystals in solution.³¹ The nanoscale size of exfoliated hBN
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28 has the potential to enable more homogeneous composites with reduced thickness in
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31 addition to providing higher surface area for interaction with the solvent/polymer system.
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35 Towards this end, we demonstrate here the fabrication of highly porous lithium-ion
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38 battery composite separators based on exfoliated hBN nanosheets and polyvinylidene
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41 fluoride (PVDF) polymers with superlative electrolyte wettability, ionic conductivity, and
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44 thermal stability. The hBN nanosheets are exfoliated using a liquid-phase process based
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48 on ethyl cellulose as a polymer stabilizer that results in a carbon coating upon thermal
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52 pyrolysis, which promotes strong interfacial interactions with the polymer matrix and
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56 enhances wettability for a range of battery electrolytes. By employing phase-inversion
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during the membrane formation process, a highly porous microstructure is achieved in which the hBN nanosheets are uniformly dispersed within the polymer matrix. Importantly, this processing method is free of plasticizers as pore formation agents and does not require additional stages of processing such as stretching, etching, or immersion in coagulation bath,²³ ultimately allowing for the direct use of the free-standing composite separator immediately after drying. The high porosity and electrolyte wettability of these phase-inversion composite separators result in enhanced electrochemical performance in LIBs including enhanced rate capability and cyclability in comparison to conventional polyolefin separators. Furthermore, the high hBN content in these composite separators leads to exceptional thermal stability, particularly enabling safe LIB operation at a high temperature of 120 °C. Overall, this work will expedite ongoing efforts to improve the safety and elevated temperature operation of rechargeable LIB technology.

Experimental Section

Hexagonal Boron Nitride Exfoliation, Processing, and Characterization. 120 g of bulk hexagonal boron nitride (hBN, Sigma-Aldrich) and 12 g of ethyl cellulose (EC, viscosity 4

cP, Sigma-Aldrich) were dispersed in 800 mL ethanol (200 proof, Decon Labs) as previously described.³² The solution was then placed in a laboratory shear mixer (Silverson L5M-A) for 2 h at 10,230 rpm. A centrifugation step at 4,000 rpm for 20 min (Beckman Coulter Avanti J-26 XPI centrifuge, JS 7.5 rotor) was followed to crash out large and unexfoliated particles. The supernatant was withdrawn and subjected to a flocculation step with a 0.04 mg mL⁻¹ sodium chloride solution to remove excess EC. The hBN/EC nanosheets were then isolated by centrifugation for 6 min at 7,500 rpm, washed with water, and dried at room temperature overnight. The exfoliated hBN/EC powder was subsequently heated in a box furnace at 400 °C for 2 h to decompose the EC and form a carbonaceous coating on the surfaces of the exfoliated hBN nanosheets.

hBN Nanosheet Composite Separator Fabrication. The hBN composite separators were prepared using dry phase-inversion. In particular, 0.17 g of previously exfoliated and annealed hBN/EC powder was dispersed in 2.85 mL of *N*-methyl-2-pyrrolidone (NMP, anhydrous, Sigma-Aldrich) by bath sonication for 2 – 4 h to form a homogeneous mixture. Next, 0.17 g polyvinylidene fluoride (PVDF, battery binder grade, MTI Corporation) and 0.15 mL of glycerol were added to the mixture, obtaining a viscous hBN/PVDF solution

with total solid content of 50 wt% hBN and 50 wt% PVDF in 3.0 mL of NMP/glycerol (95/5 w/w solvents). The mixture was stirred at 50 °C in a silicone oil bath overnight and allowed to settle for an additional 12 h to avoid bubble formation. The homogeneous hBN/PVDF solution was then blade-coated onto clean, dry, and smooth glass plates using a micrometer adjustable film applicator (EQ-Se-KTQ-50). The as-casted hBN/PVDF films were immediately transferred to an oven at 90 °C overnight to remove NMP, followed by annealing at 120 °C under vacuum for 12 h for glycerol removal. The hBN films were then punched into 19 mm diameter discs and used to assemble lithium-ion batteries.

Exfoliated hBN/EC and hBN Nanosheet Composite Separator Characterization.

Transmission electron microscopy (TEM, JEOL JEM-2100, 200 kV) was carried out to confirm the pyrolyzed EC carbon coating on the surface of the hBN nanosheets. The thermal behavior of the exfoliated hBN/EC and hBN nanosheet composite separator was characterized by thermogravimetric analysis (Mettler Toledo TGA/SDTA851 analyzer) between 25 °C and 600 °C, at a heating rate of 10 °C min⁻¹ and synthetic air flow rate of 50 mL min⁻¹ (Supporting Information, Figure S1). X-ray photoelectron spectroscopy (XPS) analysis was performed on an ESCALAB 250 XI⁺ instrument (Thermo Fisher

Scientific) employing charge compensation during measurements. XPS data analysis were performed using the Thermo Scientific Avantage data system. The porous microstructure and cross-section of the hBN nanosheet composite separator were examined by scanning electron microscopy (SEM) imaging using a Hitachi SU8030 microscope. The porosity of the separators was determined via the *n*-butanol soaking method by weighing separator discs (diameter = 1.0 cm) before and after absorption of *n*-butanol (Sigma-Aldrich) for 1 h.^{32,33} The porosity was calculated using Equation 1:

$$Porosity (\%) = \frac{w_1 - w_0}{\rho \times V} \times 100\% \quad (1)$$

where w_0 and w_1 are the weight of dry separator and after *n*-butanol absorption, respectively; ρ is the density of *n*-butanol; and V is the geometric volume of the separator.

For the electrolyte uptake tests, separator discs (diameter = 1.0 cm) were immersed in lithium hexafluorophosphate in ethylene carbonate/diethyl carbonate liquid electrolyte (1.0 M LiPF₆ in EC/DEC (1/1 v/v), Sigma-Aldrich) for 2 h. Excessive electrolyte was removed using a Kimtech wipe, followed by weighing of each separator disc. The electrolyte uptake was calculated using Equation 2:

$$\text{Electrolyte uptake (\%)} = \frac{w_1 - w_0}{w_0} \times 100\% \quad (2)$$

where w_0 and w_1 are the weight of dry separator and after electrolyte retention, respectively.

The wetting properties were evaluated by contact angle measurements (VCA Optima XE), where 3 μ L drops of different liquid electrolytes including 1.0 M LiPF₆ in EC/DEC (1/1 v/v), 0.5 M lithium bis(oxalate)borate (LiBOB, Sigma-Aldrich) in propylene carbonate (PC, Sigma-Aldrich), and 1.0 M lithium bis(trifluoromethanesulfonyl)imide (LiTFSI, Sigma-Aldrich) in 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (EMIM-TFSI, Sigma-Aldrich) were deposited on the separator surfaces. Droplet images were recorded after 1 s and 20 s, and the contact angles were measured by the sessile drop technique. The mechanical properties of the hBN separator were assessed by tensile tests in a Dynamic Mechanical Analyzer (TA Instruments RSA G2) with strain rate of 5.0 %/min in ambient conditions (ASTM D882) using rectangular hBN separator samples with dimensions of 3 mm x 10 mm.

Electrochemical impedance spectroscopy (EIS) measurements were employed to calculate the ionic conductivity of the hBN separators. Coin cells consisting of two parallel stainless-steel disk electrodes (area = 1.86 cm²) sandwiching the separator layer embedded with liquid electrolyte (1.0 M LiPF₆ in EC/DEC (1/1 v/v)) were aged overnight prior to the EIS measurements using a potentiostat/galvanostat/EIS (SP-300 Bio-Logic Science Instruments), applying an AC amplitude of 10 mV over a frequency range from 0.1 Hz to 1 MHz. The temperature of EIS measurements was precisely controlled from -10 °C to 60 °C using an environmental chamber (ESPEC BTL-433). The coin cells were equilibrated for 30 min at each temperature prior to recording measurements. The ionic conductivity (σ) was calculated using Equation 3:

$$\sigma = \frac{t}{R \times A} \quad (3)$$

where t corresponds to the separator thickness; R is the resistance determined from EIS; and A is the area of the stainless-steel disk electrode.

Linear sweep voltammetry was used to examine the electrochemical stability of the separators (Supporting Information, Figure S2). Coin cells comprised of a stainless-steel

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3 disk as a working electrode, a separator filled with liquid electrolyte (1.0 M LiPF₆ in
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7 EC/DEC (1/1 v/v)), and lithium metal foil as a counter/reference electrode were swept in
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10 a potentiostat/galvanostat (SP-300 Bio-Logic Science Instruments) with scan rate of 1.0
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13 mV s⁻¹. The thermal stability of the separators was examined by keeping the samples on
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17 a hot plate for 5 min at increasing temperatures up to 130 °C. Photographs were taken
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20 after each temperature exposure. Commercial polyolefin-based separators composed by
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23 polypropylene/polyethylene/polypropylene (tri-layer PP/PE/PP, Celgard 2325) and
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27 polyethylene (single-layer PE, Asahi Kasei) were used as sample controls in the
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31 separator property characterization.
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35 **Lithium-Ion Battery Measurements.** Lithium titanate spinel (LTO) electrodes were
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38 fabricated by coating a slurry containing 80% active material (Li₄Ti₅O₁₂ nanopowder,
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41 Sigma-Aldrich), 10% carbon black, and 10% PVDF binder on aluminum foil as current
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44 collector. LTO|Li half-cells were assembled using the hBN nanosheet composite
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47 separator filled with 1.0 M LiPF₆ in EC/DEC (1/1 v/v) liquid electrolyte and cycled on a
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51 battery tester equipment (BT-2143 Arbin Instruments). Control cells contained Celgard
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55 2325 as the separator instead. Rate capability tests were performed at 0.1C, 0.5C, 2.5C,
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5C, and 0.1C for 5 cycles at each current rate and a voltage range of 1.0–2.5 V. Cycle life tests were performed at a current rate of 0.5C for 100 cycles. For high-temperature battery measurements, the liquid electrolyte was replaced by 1.0 M LiTFSI in EMIM-TFSI ionic liquid. Single-layer PE separators (Asahi Kasei) were used for assembling control cells. LTO|Li half-cells were cycled at a constant 0.2C current rate for 15 cycles inside an environmental chamber (ESPEC BTL-433) at 120 °C. All battery cells were assembled in a glove box filled with argon gas and aged overnight prior to testing.

Results and Discussion

hBN Nanosheet Composite Separator Fabrication. hBN nanosheets were exfoliated by shear-mixing bulk hBN in ethanol using ethyl cellulose (EC) as a polymer stabilizer (see Experimental Section for details). The resulting hBN nanosheet/EC powder was then thermally annealed to pyrolyze the EC, which promoted a carbonaceous coating on the surface of the hBN nanosheets (Supporting Information, Figure S3). The fabrication of hBN nanosheet composite separators was achieved by dry phase-inversion of hBN nanosheets in polyvinylidene fluoride (PVDF), resulting in flexible (Supporting

Information, Figure S4a), porous, free-standing separators, as illustrated in the schematic drawing in Figure 1. In particular, the carbon-coated hBN nanosheets were dispersed in *N*-methyl pyrrolidone (NMP) as a solvent followed by the addition of PVDF polymer and glycerol to act as a non-solvent (Figure 1a). The hBN/PVDF solution was then blade-coated on a glass substrate and subsequently placed in an oven at 90 °C overnight to evaporate the solvent, causing the film composition to have a higher non-solvent and hBN/PVDF content. The non-solvent was then removed at 120 °C under vacuum, leaving behind a porous hBN/PVDF composite film (Figure 1b). The optimized composition was a 1:1 ratio of hBN:PVDF, resulting in composite films with suitable porosity, thickness, and mechanical flexibility for application as a LIB separator. Reducing or increasing the hBN:PVDF ratio beyond the optimal 1:1 ratio resulted in inferior ionic conductivity or mechanical durability as shown in Figure S5 (Supporting Information).

Figure 1c shows top-view and cross-sectional SEM images of the hBN separator, which exhibits a highly interconnected porous microstructure both on the surface and along the entire film thickness. Higher magnification SEM images reveal that the hBN nanosheets are homogeneously dispersed throughout the polymer matrix (Supporting Information,

Figure S6). An hBN composite separator made from bulk, non-exfoliated hBN was also prepared as a control, which showed a denser surface morphology with fewer pores (Supporting Information, Figure S7a). Furthermore, aggregated hBN particles are observed throughout the separator thickness for the control sample (Supporting Information, Figure S7b,c), resulting in less homogeneous and thicker membranes. For additional comparison, the surface morphology of commercially available separators (trilayer polypropylene/polyethylene/polypropylene – PP/PE/PP and single-layer polyethylene – PE) is provided in Figure S8 (Supporting Information).

The exfoliated hBN nanosheets play a critical role in the final porous microstructure. The dry phase-inversion mechanism begins with the initial evaporation of NMP from the as-deposited hBN/PVDF film. Concurrently, a concentration gradient of hBN/PVDF is formed between the NMP and the glycerol, creating an exchange between the solvent and non-solvent.³⁴ Therefore, the formation of the microporous structure is a result of the drying mechanism during the phase-inversion process and the strong interaction between the non-solvent and the hBN nanosheets.²³ Specifically, the carbonaceous coating on the hBN nanosheet surface leads to a favorable interaction between the glycerol and hBN

nanosheets through hydrogen bonding (Hansen hydrogen bonding parameter of glycerol = 29.3 (MPa)^{1/2}),³⁵ resulting in both a slower non-solvent evaporation rate and the hBN nanosheets and PVDF occupying the glycerol phase. Consequently, the final composite film possesses a more homogeneous network of microvoids following the complete removal of the non-solvent. For comparison, pristine PVDF separators were prepared by the identical phase-inversion method. These pristine PVDF separators showed a denser microstructure with fewer and irregular pores on the surface and along the thickness, reinforcing the importance of the hBN nanosheets in the separator composition (Supporting Information, Figure S9).

hBN Nanosheet Composite Separator Properties. The hBN nanosheet composite separator has a porosity of 59%, whereas the bulk hBN separator possesses a lower porosity of 44%, as shown in Table S1 (Supporting Information). Commercially available separators (trilayer PP/PE/PP and single-layer PE) exhibited even lower porosity values of 38% and 41%, respectively (Table S1). Porosity is crucial to entrapping a sufficient amount of electrolyte, which facilitates lithium-ion transport and enables high ionic conductivity. In general, the amount of electrolyte uptake is strongly correlated to the film

porosity (Table S1). Specifically, the separator prepared using bulk hBN showed electrolyte uptake of 224%, which increased to 353% for the separator containing the carbon-coated, exfoliated hBN nanosheets.

Figure 2a shows the ionic conductivity as a function of temperature for the hBN nanosheet composite separator and the commercial PP/PE/PP separator infiltrated with LiPF₆ in EC/DEC liquid electrolyte. At room temperature, the hBN nanosheet composite separator showed an ionic conductivity value of 0.95 mS cm⁻¹, outperforming the polyolefin-based separator by almost a factor of two. The ionic conductivity also demonstrated a strong temperature dependence, resulting in ionic conductivities at 60 °C of 1.7 mS cm⁻¹ and 0.93 mS cm⁻¹ for the hBN nanosheet composite separator and PP/PE/PP separator, respectively. The lowest ionic conductivity values for both separators were observed at -10 °C. However, the hBN separator still outperforms the commercial polymer membrane, yielding an ionic conductivity of 0.50 mS cm⁻¹, which is sufficient for typical LIB applications.

Ionic conductivity depends on the concentration of ions and their mobility.³⁶ The higher ionic conductivity for the hBN nanosheet composite separator at all measured

temperatures in comparison to the commercial PP/PE/PP separator is thus consistent with the high porosity provided by the phase-inversion process. In addition to porosity, the increased ionic conductivity of the hBN nanosheet composite separator can be ascribed to its improved electrolyte wettability. The wetting of liquid electrolytes was observed by applying a droplet of different liquid electrolytes on the separator surface (Figure 2b). The hBN nanosheet composite separator showed enhanced wettability for a variety of common electrolytes (e.g., LiPF_6 EC/DEC, LiBOB/PC, and LiTFSI/EMIM-TFSI) as confirmed by contact angle measurements (Figure 2c,d). In particular, the LiPF_6 EC/DEC electrolyte was readily absorbed by the hBN nanosheet composite separator, resulting in complete wetting within 20 s, whereas the commercial PP/PE/PP membrane showed decreased electrolyte absorption with considerably higher contact angle. The same trend was observed for the LiBOB/PC electrolyte, which rapidly infiltrated the hBN nanosheet composite separator, resulting in a low contact angle of 21° . In contrast, the polyolefin membrane showed a higher contact angle of 71° that did not decline after 20 s. The enhanced wettability of the hBN nanosheet composite separator is consistent with the expected favorable interaction between the carbon-coated hBN nanosheets and the

carbonate-based electrolyte solvents. In addition, the highly polar C–F covalent bonds in the polymer matrix reinforces the electrolyte wettability.³⁷ The inferior wettability of the PP/PE/PP separators is attributable to the significant difference in polarity between the polyolefins and the electrolyte.^{38,39} Relatively poor electrolyte wettability was also observed for the single-layer PE separator (Supporting Information, Figure S10).

The wettability of the separators towards ionic liquid electrolytes was also examined. The hBN nanosheet composite separator was effectively infiltrated with LiTFSI/EMIM-TFSI ionic liquid electrolyte, showing a contact angle of 30° that dropped to 21° within 20 s. The wetting properties of LiTFSI/EMIM-TFSI ionic liquid are influenced by the large polarity and non-polarity of the ion pair. In this case, the TFSI⁻ anion is an amphiphile possessing both strong hydrophobic and hydrophilic moieties, and is capable of forming hydrogen bonds with the EMIM⁺ cation.⁴⁰⁻⁴² Therefore, the wetting properties of the hBN nanosheet separators towards LiTFSI/EMIM-TFSI are mostly correlated to the favorable interactions of the electrolyte with the carbon-coated hBN nanosheets and the polar PVDF polymer. As the PP/PE/PP membrane is comprised of nonpolar polyolefins, this

separator showed much weaker wettability for LiTFSI/EMIM-TFSI, with a constant contact angle value of 68° that did not decline with time.

Since the carbon coating on the hBN nanosheets is critical to the formation and wettability of the composite separator, further analysis was performed to elucidate this surface chemistry as shown in Figure 3. Figure 3a shows the nanoscale morphology of the carbon-coated hBN nanosheets as observed by SEM, which matches the morphology of the hBN nanosheets on the surface of the hBN nanosheet composite separator (Figure 1c). The surface characterization of the carbon-coated hBN nanosheets and the hBN nanosheet composite separator was investigated by XPS analysis, as shown in Figure 3b-d. The XPS survey spectra (Figure 3b) identified all elements comprising both sample surfaces. Peaks assigned to boron, nitrogen, carbon, and oxygen are present in both samples, whereas an additional fluorine peak at ~ 688 eV is observed for the hBN nanosheet composite separator due to the presence of fluorine in PVDF. The quantitative compositional analysis was calculated from the XPS survey spectra with the atomic percentages of boron, nitrogen, carbon, oxygen, and fluorine being provided in Figure 3c. Both samples exhibited spectral features consistent with the presence of pyrolyzed EC

carbon coating on the surface of the hBN nanosheets, which are exclusively composed of C and O. The hBN nanosheet composite separator showed suppressed atomic percentages of B and N owing to the blend with the PVDF polymer, which in turn resulted in higher concentration of C and F. The relative atomic percentage of O is approximately constant, reflecting the presence of oxygen-containing moieties in the carbon coating on the hBN nanosheets.

The C 1s spectrum of the carbon-coated hBN nanosheets indicates carbon in different chemical states with three deconvoluted components: C–C, C–O, and O–C=O, with binding energy peaks at ~285 eV, ~286 eV, and ~289 eV, respectively (Figure 3d). The C 1s spectrum of the hBN nanosheet composite separator was fitted considering four different contributions that can be associated with C–C (~285 eV), CH₂–CF₂/C–O (~287 eV), O–C=O (~288 eV), and CF₂ (291 eV).⁴³ This spectral fitting confirmed that the coating of the hBN nanosheets consists of carbonaceous species composed of C–C and oxygen-containing groups (O1s XPS spectra are shown in Figure S11, Supporting Information), which promote favorable electrolyte interactions that ultimately enhance electrochemical properties as will be delineated below.

The mechanical properties of the hBN separator were also assessed and compared to previously reported separators that have been successfully employed in LIBs. Figure S4b presents the stress-strain curve of the hBN separator, showing a tensile strength of 5.3 MPa. This value is consistent with PVDF-based composites that have been used as LIB separators.^{23,44-46} For instance, the hBN separator showed comparable tensile strength to the PVDF/Al₂O₃ composite separator reported by Blake *et al.* (5.4 MPa).²³ In addition, the hBN separator displayed a high elongation of 60.3%, whereas PVDF/Al₂O₃ composite separators exhibited a lower elongation of 28%, thus confirming that the hBN separator has suitable mechanical properties for LIB applications.

hBN Nanosheet Composite Separator Battery Testing. To assess electrochemical performance, the separators were incorporated into LIBs in a half-cell configuration using Li₄Ti₅O₁₂ (LTO) electrodes and LiPF₆ in EC/DEC liquid electrolyte. Figure 4a shows the rate capability tests from 0.1C to 5C for cells made with the hBN nanosheet composite separator and control cells assembled using the commercial PP/PE/PP separator. At various applied current densities, both cells exhibited stable cycling. However, in accordance with its higher ionic conductivity, the hBN nanosheet composite separator

yielded a higher capacity at all current rates compared to the control cell using the commercial PP/PE/PP separator. The corresponding charge-discharge voltage profiles in Figure 4b showed capacities of 168 mAh g⁻¹ and 162 mAh g⁻¹ at 0.1C for the hBN nanosheet composite separator and PP/PE/PP membrane, respectively. At high rate, where ionic conductivity plays a more limiting role, the hBN nanosheet composite separator extends its advantage over the PP/PE/PP membrane, particularly showing a 5C capacity of 132 mAh g⁻¹ compared to 120 mAh g⁻¹. The LTO half-cells were also subjected to cycle life testing at 0.5C as shown in Figure 4c. The hBN nanosheet composite separator performance exceeded the commercial PP/PE/PP separator during the test, revealing high stability and retained capacity of 158 mAh g⁻¹ after 100 cycles. The corresponding voltage profiles in Figure 4d illustrate the stable charge-discharge behavior and minimal capacity fading after extended cycling.

The thermal stability behavior of the hBN nanosheet composite separator and the commercial PE separator is shown in Figure 5a, where both samples were subjected to identical heating conditions. It is evident that the hBN nanosheet composite separator has excellent thermal stability up to 130 °C with no shrinkage or other observable changes,

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4 whereas the commercial PE separator undergoes irreversible loss of structural integrity
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7 that becomes more noticeable with increasing temperature. Equivalent morphological
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10 degradation upon heating was seen for the commercial PP/PE/PP separator (Figure S12,
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13 Supporting Information). The improved thermal stability of the hBN separator is
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16 attributable to the refractory nature of hBN nanosheets,⁴⁷ which also act as promising
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19 flame retardant nanoadditives for polymers.⁴⁸ Indeed, the hBN nanosheet composite
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22 separators were also stable upon exposure to a flame, whereas the polyolefin-based
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25 separators were also stable upon exposure to a flame, whereas the polyolefin-based
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28 commercial separators burned and completely lost structural integrity as shown in Figure
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31 S13 (Supporting Information).
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35 Conventional liquid electrolytes utilizing LiPF_6 as the conducting salt dissolved in liquid
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38 carbonate-based electrolytes suffer from poor thermal stability due to hydrolyzation and
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41 degradation of salt and/or solvent components, which results in side reactions that
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44 compromise electrodes and induce severe capacity fade.⁴⁹ Consequently, thermally
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47 stable LiTFSI/EMIM-TFSI ionic liquid electrolytes were used for high-temperature battery
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50 testing. In particular, the LiTFSI salt presents higher thermal stability and low sensitivity
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53 to hydrolyzation while the EMIM-TFSI ionic liquid overcomes safety concerns due to
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negligible vapor pressure and nonflammability.⁵⁰⁻⁵² Since this ionic liquid electrolyte did not wet the PP/PE/PP separator (Figure 2d), the control cell in this case was assembled using the PE separator.

Figure 5b and Figure 5c show the cycling stability and corresponding voltage profiles for LTO half-cells using the hBN nanosheet composite separator and PE separator at 0.2C and an operating temperature of 120 °C. The high-temperature cycling stability of the hBN nanosheet composite separator clearly exceeds the PE separator with more than double the capacity retention after 15 cycles. These results underscore the potential of the hBN nanosheet composite separators to enhance LIB safety in addition to enabling high-temperature applications for LIBs such as autoclavable surgical tools and subsurface exploration devices in addition to aerospace and military technologies.

Conclusions

In summary, we have demonstrated the fabrication of free-standing LIB composite separators based on hBN nanosheets prepared using dry phase-inversion. The scalable liquid-phase exfoliation of bulk hBN into hBN nanosheets was performed in the presence

of EC as a polymer stabilizer. Subsequent EC pyrolysis resulted in a nanoscale carbonaceous coating on the surface of the hBN nanosheets that facilitated favorable interactions with the PVDF polymer matrix and promoted electrolyte wettability. The resulting high porosity hBN nanosheet composite membranes possessed high ionic conductivity that enhanced LIB rate capability and cycling stability compared to conventional polyolefin separators. Moreover, the high thermal stability of the hBN nanosheet composite separators allowed effective LIB operation at an extreme temperature of 120 °C. Overall, this work establishes a promising strategy for producing LIB separators that enhance electrochemical performance, improve safety, and extend operating temperature range.

Associated Content

Supporting Information. The Supporting Information is available free of charge on the ACS Publications website.

Supporting characterization data including transmission electron microscopy images, scanning electron microscopy images, contact angle measurements, X-ray photoelectron

spectra, electrochemical stability curves, thermogravimetric analysis, and flame combustion testing (PDF).

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Author Contributions

A. C. M. M. and W. J. H. contributed equally to this work. All authors discussed the results and contributed to the writing and editing of the manuscript. All authors have given approval to the final version of the manuscript.

Notes

The hBN nanosheet composite separator is being commercialized in collaboration with MilliporeSigma.

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Figures



Figure 1. Fabrication of the hBN nanosheet composite separator. (a) The hBN/PVDF solution is prepared by dispersing the previously annealed EC-exfoliated hBN nanosheet powder and PVDF polymer in a 95:5 ratio of *N*-methyl-2-pyrrolidone and glycerol. (b) The hBN/PVDF solution is then blade-coated on a glass substrate followed by drying to

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4 promote phase-inversion by sequential removal of the solvent and non-solvent. The hBN
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7 nanosheet composite precipitates during this process, ultimately forming a porous and
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10 free-standing film. (c) Top-view and cross-sectional SEM images of the as-prepared hBN
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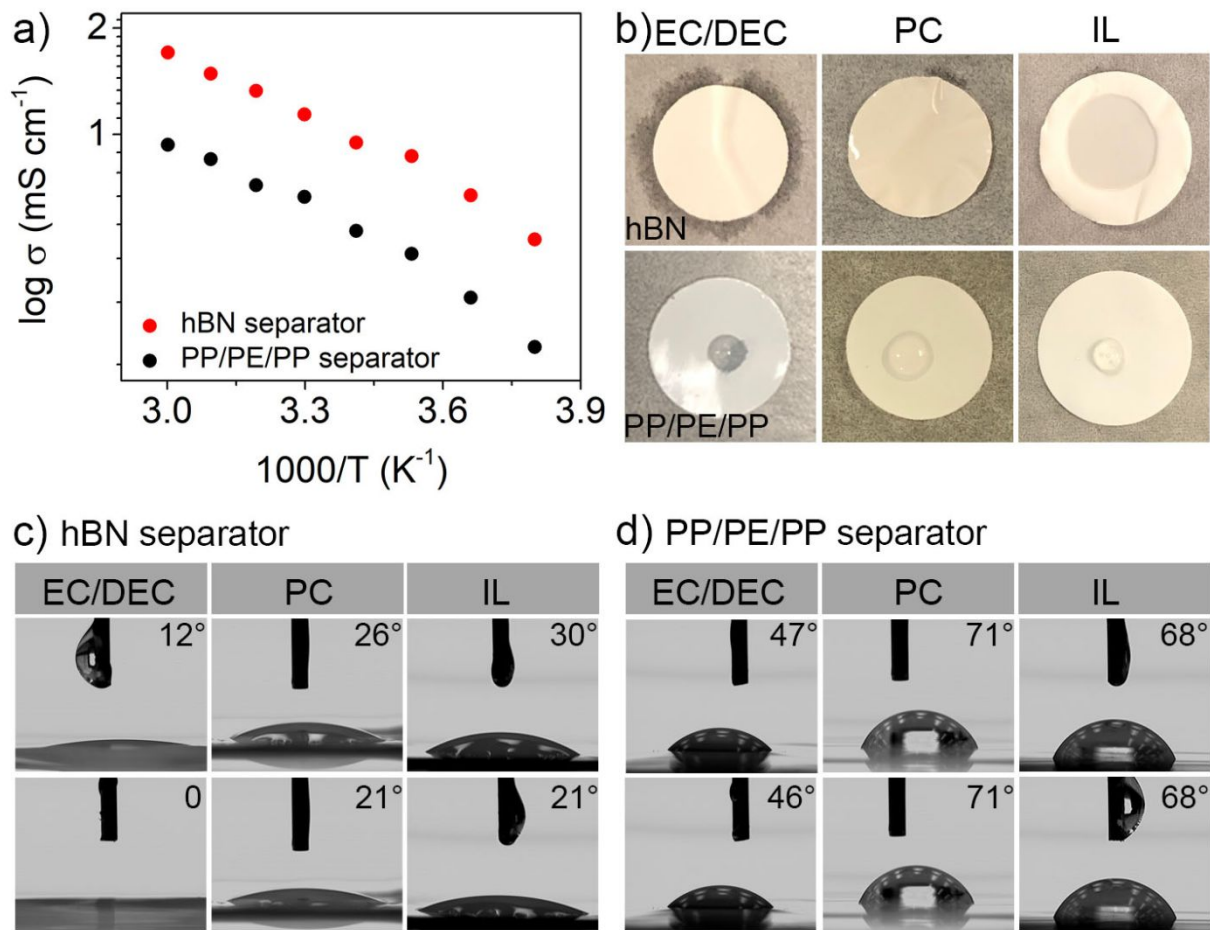


Figure 2. (a) Ionic conductivity as a function of temperature for the hBN nanosheet composite separator compared to a commercial PP/PE/PP separator. (b-d) Wetting properties of the hBN nanosheet composite separator compared to a commercial PP/PE/PP separator for various liquid electrolytes (EC/DEC = LiPF₆ in EC/DEC (1/1 v/v); PC = LiBOB/PC; IL = LiTFSI in EMIM-TFSI ionic liquid). (b) Photographs of liquid electrolyte droplets deposited on the surface of separators were taken after exposure for

3 min. Contact angle images for (c) hBN nanosheet composite separator and (d) commercial PP/PE/PP separator. Photographs were taken after 1 s (upper row) and 20 s (lower row).

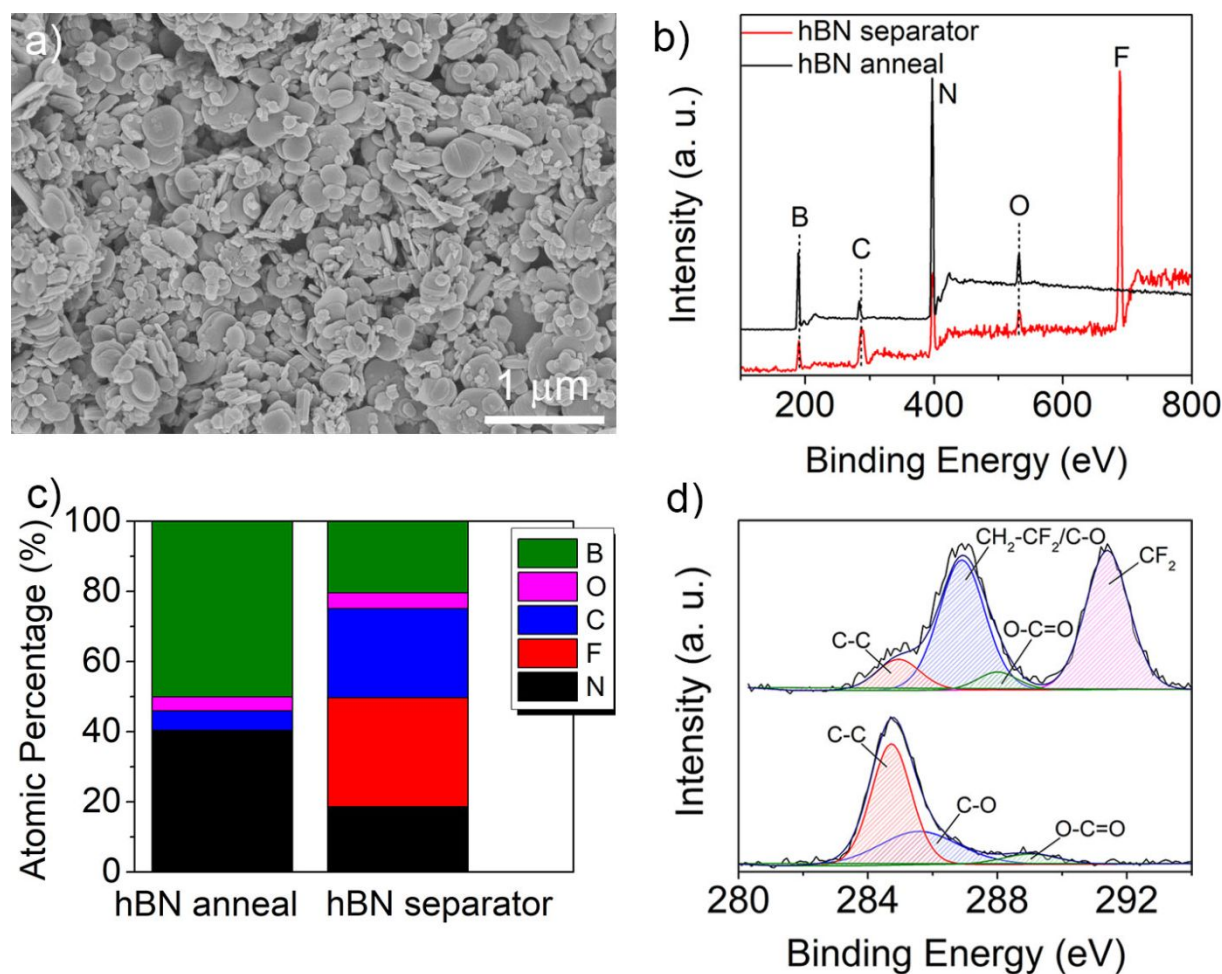


Figure 3. Annealed EC-exfoliated hBN nanosheet powder and hBN nanosheet composite separator surface properties. (a) SEM image of the annealed hBN/EC powder. (b) XPS survey spectra of the annealed hBN/EC powder and the hBN nanosheet composite separator. (c) Atomic percentage of the annealed hBN/EC powder and the hBN nanosheet composite separator. (d) C 1s XPS spectra for the annealed hBN/EC powder (lower plot) and the hBN nanosheet composite separator (upper plot).

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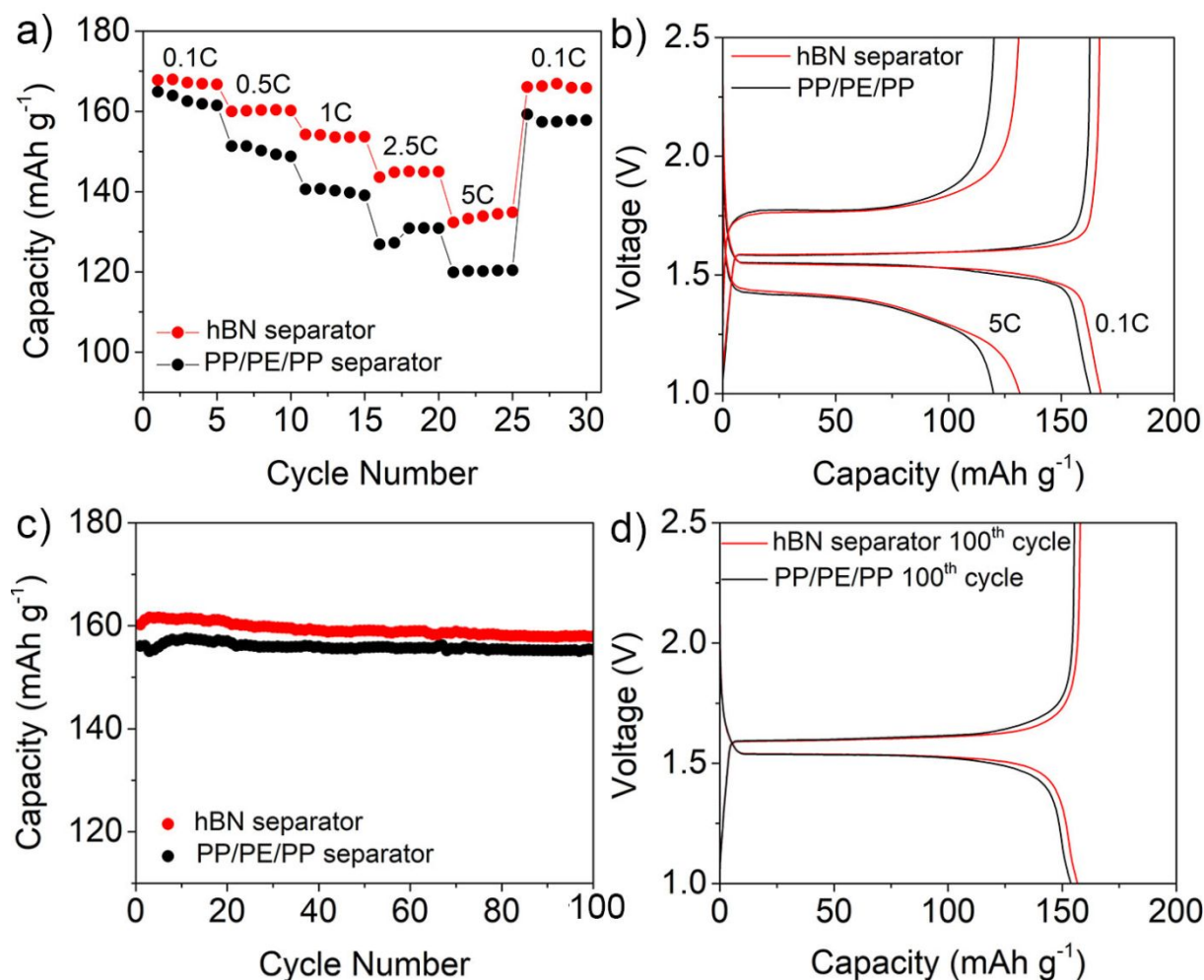


Figure 4. Electrochemical characterization of the hBN nanosheet composite separator and the commercial PP/PE/PP separator using lithium titanate (LTO) anode electrodes and 1.0 M LiPF₆ in EC/DEC (1/1 v/v) liquid electrolyte at room temperature. (a) Rate capability performance tests from 0.1C to 5C. (b) Charge-discharge voltage profiles at 0.1C and 5C rate. (c) Cycle life performance tests at 0.5C. (d) Charge-discharge voltage profiles after 100 cycles at 0.5C.

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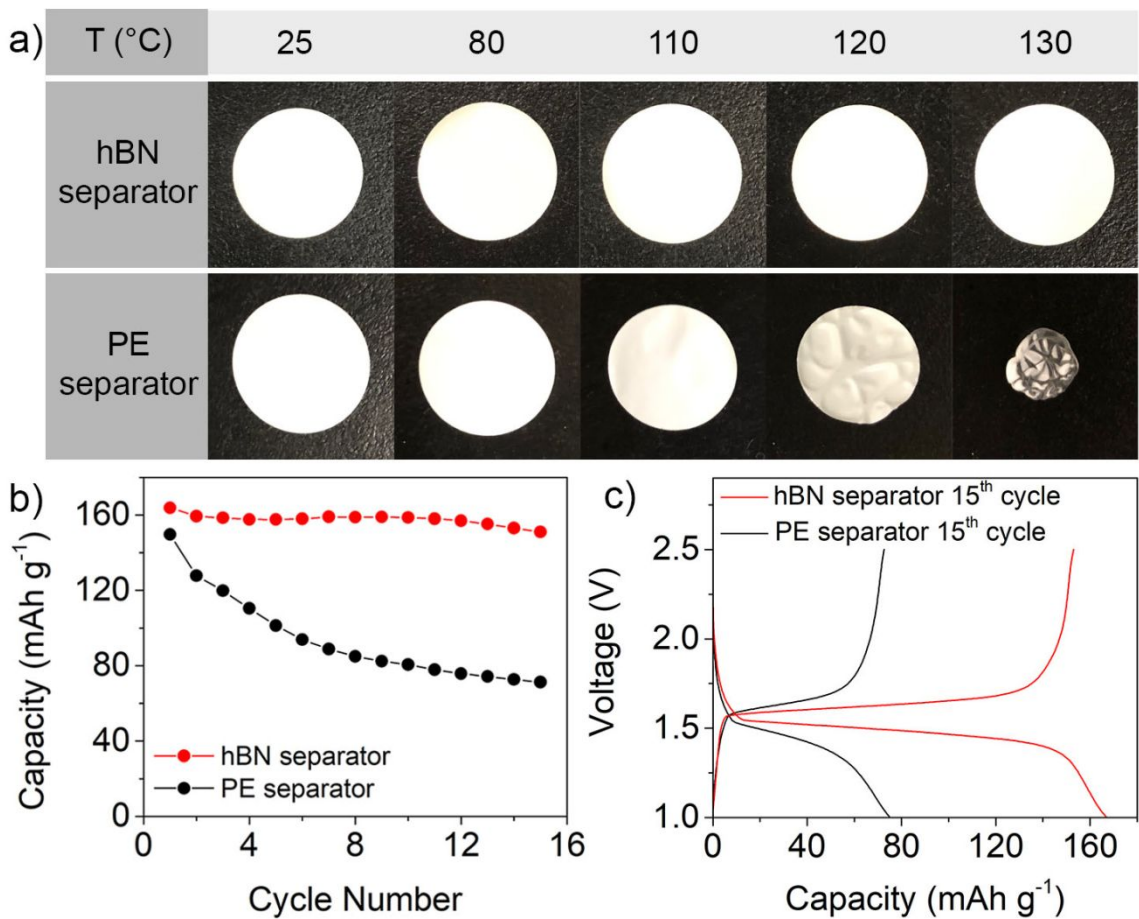
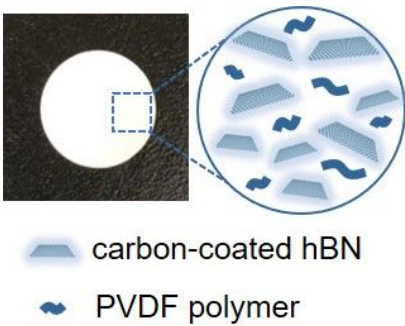


Figure 5. Physical and electrochemical characterization of the hBN nanosheet composite separator and commercial PE separator at increasing temperatures. (a) Thermal properties of the hBN nanosheet composite separator and commercial PE separator. Each photograph was taken following heating for 5 min at the specified temperatures. (b) Cycling performance tests and (c) voltage profiles for LTO|Li half-cells cycled at 0.2C and 120 °C using the hBN nanosheet composite separator and commercial PE separator with LiTFSI in EMIM-TFSI ionic liquid as the electrolyte.

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